

João Paulo Canário, Ph.D.

SWE | ML Engineer | AI Researcher

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WORK EXPERIENCE

Stone

2022 – present

Staff Software Engineer (Risk Management Platform)

- Act as a high-level technical reference for the risk management platform, defining long-term architectural strategies and ensuring consistency across multiple domains;
- Lead complex technical projects involving dependencies across multiple squads (Banking, Cards, Acquiring, etc);

Senior Software Engineer (Risk Management Platform)

- Led the technical architecture of risk solutions for critical Stone products;
- Core developer on the stream feature engine, handling real-time feature computation for over 200K events per day for fraud prevention across multiple domains
- Core developer at CORTEX (Centralized Orchestrator), which enables agnostic integration and orchestration across multiple workflows, internal and external services;

Senior Data Scientist (Card Fraud Prevention)

- Responsible for analyzing card transaction data to create rules and data modeling initiatives to amplify fraud prevention assertiveness;
- Development assist of the stream feature engine application that analyzes and computes near-real-time features for over 200K events per day.

CI&T

2021 – 2022

Senior ML Engineer

- Assist in building and growing the Python Platform for Machine Learning, Data Science, and MLOps for the Via Varejo company.

Senior Data Scientist

- Aid the creation of a machine learning model for the recommendation system for the Via Varejo company.

NeoDados Analytics

2018 – 2021

Senior Software Engineer (Machine Learning)

- Architect an async image classification system that processes 1M images/day with over 80% accuracy.
- Create an ETL workflow to ingest data for a fraud detection system processing over 500K bus card usage records per day.

Additional Experience

2007 – 2017

- **Computer Vision Engineer** at EchoFlow Engineering [2015~2017]
- **Middle SWE** at Reconcavo Institute of Technology [2008~2014]
- **Internship** at Geotecnia Lab of Polytechnic School [2007~2008]

SKILLS

Programming – Python, Golang, Kotlin, Elixir, Celery, Docker, Kubernetes, Apache Flink, FastAPI, Flask, RabbitMQ, Kafka, Shell Script, REST API, gRPC, Git, JavaScript • **Science** – Data Science, Machine Learning, Computer Vision, Deep Learning • **Data** – MLOps, SQL, MongoDB, Redis, PostgreSQL, Airflow, Azure Databricks, Apache Flink, PySpark, NumPy, Polars, Pandas, OpenCV, Scikit-Learn, Keras, TensorFlow, PyTorch

EDUCATION

PhD in Computer Science Sep 2017 – Jul 2022
Federal University of Bahia
On Deep Learning Features for Noisy Time Series Classification.
Thesis defense: July 04, 2022

MOST RECENT PUBLICATIONS

Noise-Tolerant Self-Embedded LSTM for Seismic Event Classification <i>International Workshop on Machine Learning for Signal Processing</i>	2023
Fog-DeepStream: A new approach combining LSTM and Concept Drift for data stream analytics on Fog computing <i>Internet of Things Journal.</i>	2023
A face detection ensemble to monitor the adoption of face masks inside the public transportation during the COVID-19 pandemic <i>Multimedia Tools and Applications</i>	2022
Ethics of AI: Do the Face Detection Models Act with Prejudice? <i>Brazilian Conference on Intelligent Systems</i>	2021
In-depth comparison of deep artificial neural network architectures on seismic events classification <i>Journal of Volcanology and Geothermal Research</i>	2020
Llaima volcano dataset: in-depth comparison of deep artificial neural network architectures on seismic events classification <i>Data in Brief</i>	2020
Using CNN to classify spectrograms of seismic events from Llaima volcano (Chile) <i>International Joint Conference on Neural Networks</i>	2018